

Hasnain Heickal

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EDUCATION

University of Massachusetts Amherst

MS/PhD in Computer Science and Engineering. Advisor: Dr. Andrew Lan

Massachusetts, USA

Fall, 2021 – Present

University of Dhaka

MS in Computer Science and Engineering

Dhaka, Bangladesh

2012 – 2013

CGPA - 3.85/4.00. Rank - 4th among 13 students.

University of Dhaka

B.Sc. (Hons) in Computer Science and Engineering

Dhaka, Bangladesh

2007 – 2011

CGPA - 3.63/4.00. Rank - 4th among 43 students.

WORK EXPERIENCE

Summer Intern

Educational Testing Service (ETS)

June 2024 - August 2024

NJ, USA

Worked in Automatic Speech and Language Processing (ASLP) team under Product and Innovation Development (PID).

Assistant Professor (On study leave)

Lecturer

Department of Computer Science and Engineering, University of Dhaka

February 2018 – Present

May 2014 – February 2018

Bangladesh

Courses Taught: Advanced Algorithms (MS), Programming Fundamentals, Data Structures, Algorithms, Distributed Systems, Computer Graphics, System Programming, Numerical Analysis, Multi-variable Calculus and Geometry.

Lecturer

Department of Computer Science and Engineering, Eastern University

January 2014 – May 2014

Bangladesh

RESEARCH

Research Interest

Machine Learning for Education, Natural Language Processing

Google Scholar Profile

<https://scholar.google.com/citations?user=oQc7yEAAAAAJ>

Current Research

Natural language feedback generation for programming assignments

Selected Publications

1. *Learning Code-Edit Embeddings to Model Student Debugging Behavior* **Hasnain Heickal**, Andrew Lan. AIED 2025: 91–98.
2. *Generating Feedback-Ladders for Logical Errors in Programming using Large Language Models* **Hasnain Heickal**, Andrew Lan. EDM 2024: 947–951.
3. *An Automatic Abstractive Text Summarization System* Nasid Habib Barna, **Hasnain Heickal**. DUJASE 6(2) (2022): 39–48
4. *End-to-End Speech Synthesis for Bangla with Text Normalization*. Tanzir Islam Pial, Shahreen Salim Aunti, Shabbir Ahmed, **Hasnain Heickal**. CSII 2018: 66–71
5. *Deep Learning Based Bangla Speech-to-Text Conversion*. Md. Tahsin Tausif, Sayontan Chowdhury, Md. Shiplu Hawlader, Md. Hasanuzzaman, **Hasnain Heickal**. CSII 2018: 49–54
6. *Segmentation of Heterogeneous Documents into Homogeneous Components using Morphological Operations*. Nasid Habib Barna, Tisa Islam Erana, Shabbir Ahmed, **Hasnain Heickal**. ICIS 2018: 513–518
7. *A framework for developing graphically programmable low-cost robotics kit for classroom education*. Mehedi Hasan Masum, Tanvir Shahriar Rifat, Saifuddin Md Tareeq, **Hasnain Heickal**. ICETC 2018: 22–26

8. *Computer Vision-Based Real-Time 3D Gesture Recognition Using Depth Image*. Hasnain Heickal, Tao Zhang, Md. Hasanuzzaman. International Journal of Image and Graphics 15(1) (2015)

Thesis Supervisions

1. *Automatic Evaluation of Subjective Answers*. Master's thesis of Tisa Islam Erana. Proposed a weighted bipartite matching-based semantic distance measure to group similar answers.
2. *A Hybrid Filtering-based Book Recommender System Using Users' Social Media Activities*. Master's thesis of Shahreen Selim Aunti. Proposed a Hybrid Filtering-based approach that uses users' social media activities to identify their preferences and topics of interest.
3. *Sentiment Analysis using Convolutional Neural Network with Word Embedding and SentiWordNet*. Master's thesis of Tanzir Islam Pial. Proposed a novel approach for sentiment analysis using a CNN that uses word vectors and sentiment information of individual words together.

PROGRAMMING CONTEST EXPERIENCE

Coach of Programming Contest

Department of Computer Science and Engineering, University of Dhaka

2010 – Present

Department of Computer Science and Engineering, Jahangirnagar University

September 2011 – October 2013

Two teams have qualified to ICPC World Finals under my coaching in 2017 and 2020.

Problem-setting and Judging

2013 – Present

ICPC Asia West Continent Final Contest

January, 2020

- As Judging Coordinator, coordinated with the judging team to create an original problem set. Team members included judges from Iran, India, Pakistan, Afghanistan, and Bangladesh.

ICPC Dhaka Regional

2013, 2017 – 2019

- As Judging Director on 2019, managed the whole judging team consisting 23 persons.
- As Problem-setter and Judge, created few original problems, wrote solutions, tests etc.

National Collegiate Programming Contest (NCPC)

2014 - 2015, 2017 - 2020

- As Judging Coordinator on 2018 - 2020, coordinated with the judging team to create an original problem set.
- As Problem-setter and Judge, created few original problems, wrote solutions, tests etc.

Team Leader, Bangladesh Informatics Olympiad Team

2014

International Olympiad of Informatics (IOI)

Taipei, Taiwan

Competitive Programming Achievements

- 1st position in IUPC by BGC Trust University, Bangladesh in 2012. Team - DU Angry Birds
- 1st position in NCPC by University of Dhaka, Bangladesh in 2011. Team - DU Resonance
- 1st position in IUPC by IUT, Bangladesh in 2011. Team - DU Resonance
- 5th position in ICPC Dhaka Regional by NSU, Bangladesh in 2011. Team - DU Resonance
- 4th position in ICPC Dhaka Regional by NSU, Bangladesh in 2010. Team - DU Resonance
- 6th Position in ICPC Amritapuri Regional by Amrita Vishwa Vidyapeetham, India in 2010. Team - DU Resonance
- 2nd position in NCPC by SUST, Bangladesh in 2010. Team - DU Resonance
- 4th position in ICPC Amritapuri Regional by Amrita Vishwa Vidyapeetham, India in 2009. Team - DU Dark Shadows

Online Problem Solving

- Solved more than 700 problems in Valladolid Online Judge. Profile: [hhjami](#)
- Maximum rating in Topcoder – 1692 (Yellow). Profile: [Jami.CSEDU](#)
- Maximum rating in Codeforces – 2128 (International Master). Profile: [Jami.CSEDU](#)

SKILLS

Programming Languages: C/C++, Python, Java, PHP, JavaScript, SQL, QBasic, Assembly, Lisp, Prolog.

Key Concepts: Data Structures, Algorithms, NLP, Machine Learning, Databases, Object Oriented Programming

Tools: Vim, Git, Google Cloud Platform, VS Code, Eclipse, Netbeans

Libraries: OpenGL, Weka, OpenNI, TensorFlow

Frameworks: React, Node.js, WordPress, Laravel

Operating Systems: Ubuntu, MacOS, Windows, Fedora, CentOS

EXTRA CURRICULAR ACTIVITIES

- Champion in Regional Mathematics Olympiad, Chittagong, 2006.
- Second Runner Up in National Mathematics Olympiad, Bangladesh, 2006.
- Runner Up in Inter School Debate Championship, 2002, organized by Drishty, Chittagong.
- Can play several musical instruments such as Violin, Mouth Organ, etc.